

Beyond the OpenAPI contract

Resource-specific operation discovery in IRI v2

```
"iri:submit-job": {  
  "href": "/compute/job/system-a"  
}
```

WHY	relation
WHERE	href
HOW	OpenAPI

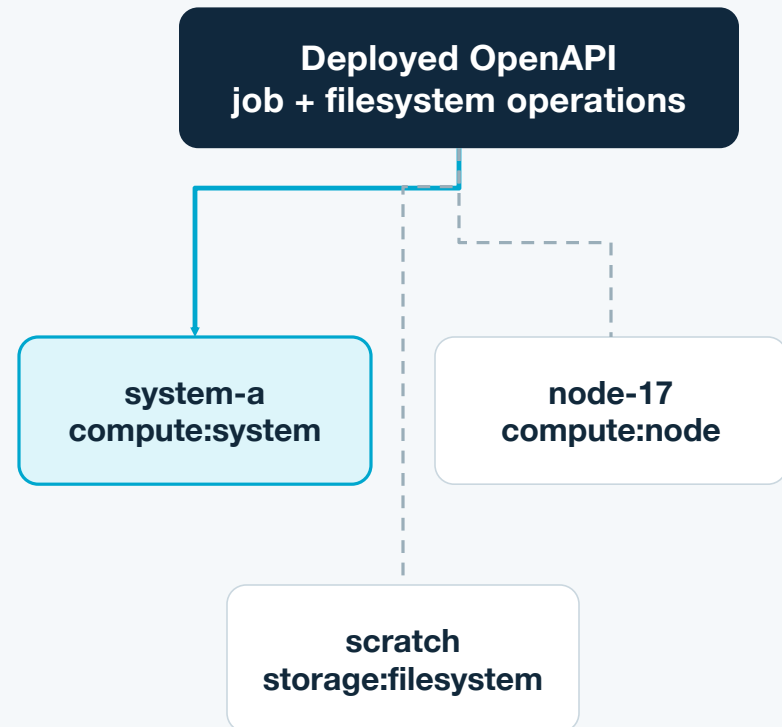
OpenAPI remains authoritative for invocation. Runtime representations supply applicability and location.

The resource-level discovery question

02

Which operations apply to system-a, and where should the client invoke them?

A deployed service contract can define job operations without selecting them for a particular Resource instance.



OpenAPI remains the invocation authority

03

The RFC preserves a single structural source of truth

```
/api/v2/compute/job/{resource_id}:  
  post:  
    operationId: launchJob  
    security: HTTPBearer  
    requestBody: JobSpec  
    responses: Job
```

METHOD + PATH

POST and deployed server resolution

INPUT CONTRACT

parameters, headers, bodies, serialization

OUTPUT CONTRACT

responses, errors, media types

SECURITY

authentication and authorization
requirements

No method or schema is duplicated in the HAL Link Object.

A complete OpenAPI still leaves runtime questions

```
POST  
/api/v2/compute/job/  
{resource_id}
```

Operation Object

01

APPLICABILITY

Does this operation apply to system-a?

02

LOCATION

Which producer-bound URI should the client use?

03

VISIBILITY

Should this caller see the operation?

04

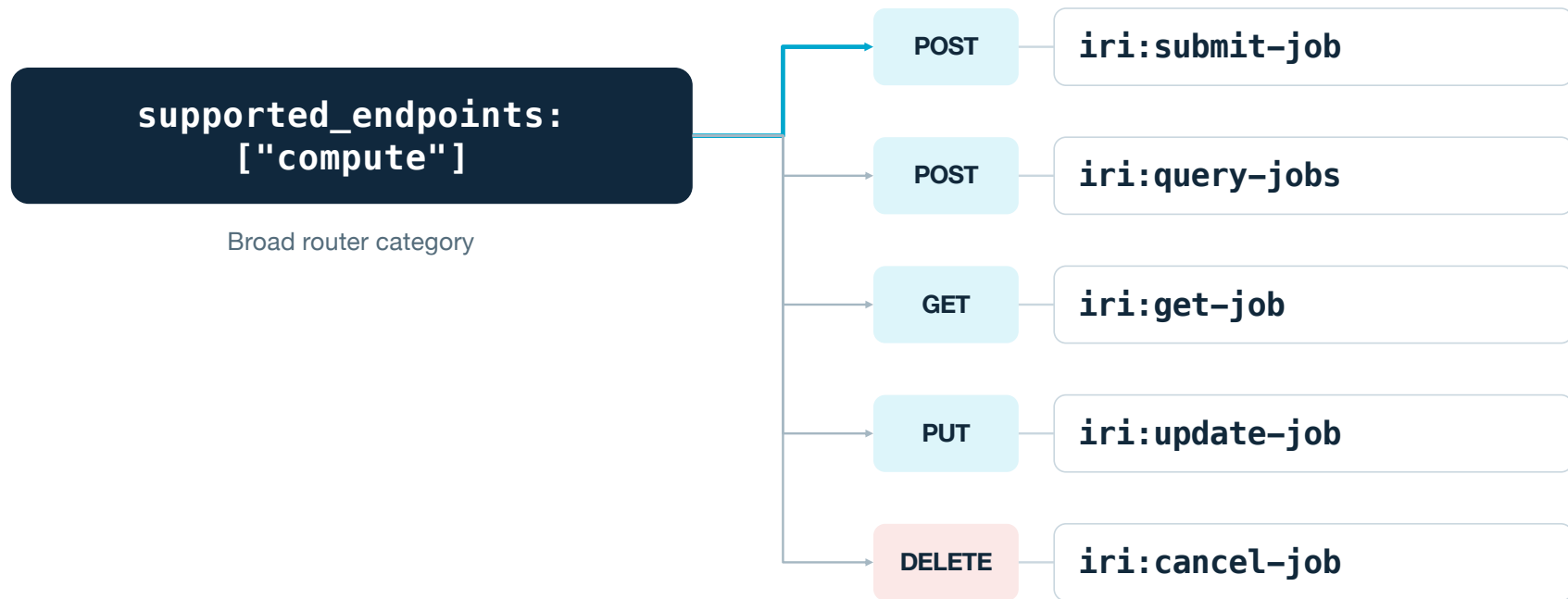
OMISSION

What does a missing link mean?

These answers may vary by Resource, adapter configuration, and caller.

One category hides five compute operations

05



The category supplies neither operation-level support nor an invocation URI.

Resource type does not establish operation support

06

Type classification narrows eligibility. Adapter context establishes applicability

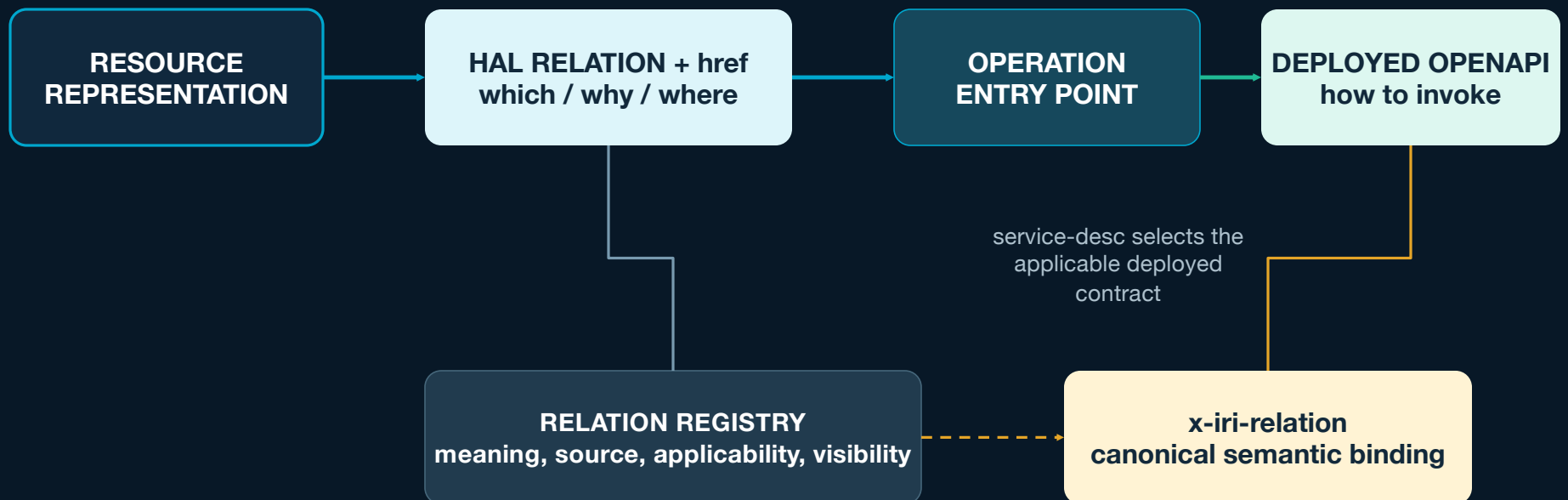
Resource type	Potential operation family	Required runtime context
compute:system	Compute and conditional filesystem	Implemented adapter and bound execution context
compute:node	Conditional filesystem only	Explicit node-specific configuration. No automatic job relations
storage:filesystem	Filesystem and storage discovery	Adapter accepts the filesystem ID and defines its path namespace
storage:mount	Conditional filesystem	Mount-specific context with no inherited operations

Type alone must not imply support.

The coordinated authority model

Each layer answers one concern and remains authoritative for it

07



A Resource advertises concrete affordances

```
{
  "id": "system-a",
  "resource_type": "...:compute:system",
  "_links": {
    "iri:submit-job": {
      "href": "/compute/job/system-a"
    },
    "iri:query-jobs": {
      "href": "/compute/status/system-a"
    },
    "service-desc": {
      "href": "/openapi.json"
    }
  }
}
```

BOUND CONTEXT

resource_id is already bound to **system-a**

SELECTIVE DISCOVERY

Only applicable operations visible to this caller appear

DEPLOYED CONTRACT

service-desc identifies the OpenAPI that governs invocation

x-iri-relation joins discovery to invocation

HAL

```
"iri:submit-job": {  
  "href": "/compute/job/system-  
a"  
}
```

OPENAPI

```
post:  
  operationId: launchJob  
  x-iri-relation:  
    - https://iri.science/rels/submit-job
```

expand CURIE

<https://iri.science/rels/submit-job>

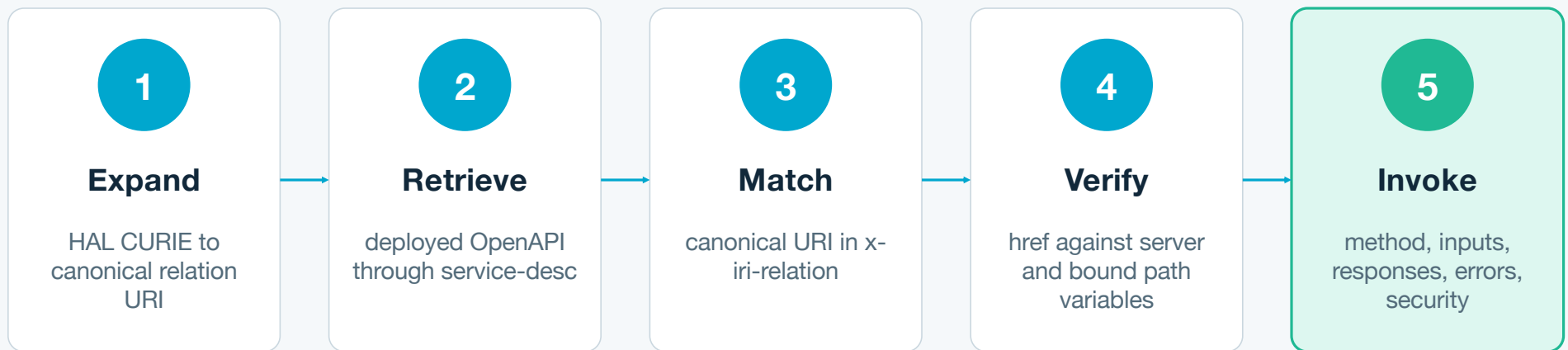
match extension

Stable binding key

operationId and deployed paths may change without renaming the relation

Client resolution and failure behavior

10



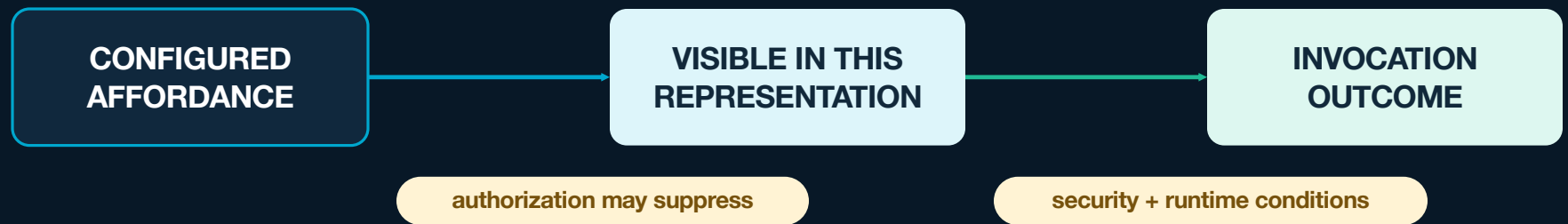
Stop on a missing, duplicate, or target-inconsistent binding.

Clients must not guess a method or probe a mutation to discover its contract.

Authorization, availability, and omission

11

Discovery and invocation remain separate decisions



PRESENCE

Configured and visible. It grants no permission and promises no success.

ABSENCE

Not advertised in this representation. It does not prove global lack of support.

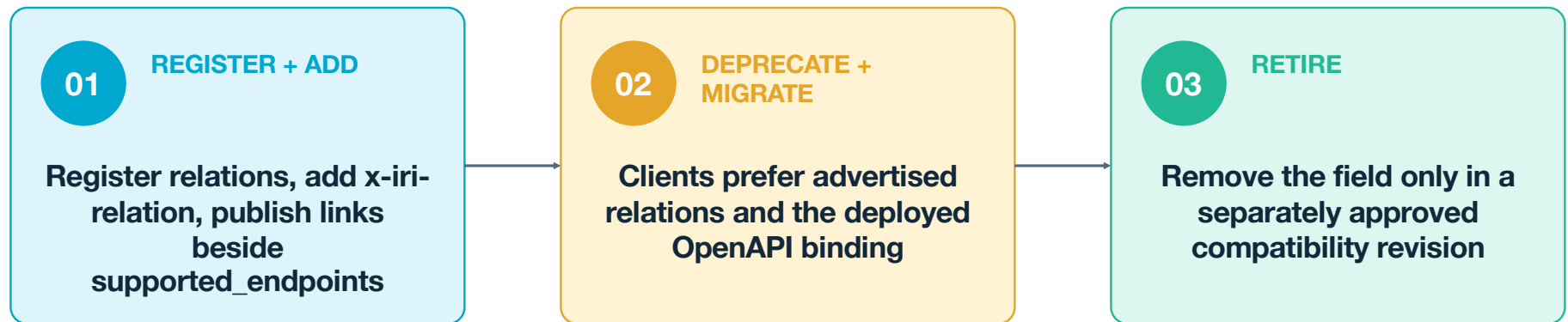
FAILURE

Security, capacity, queue state, or an outage may still reject the invocation.

Temporary outage alone should not remove a configured Resource operation link.

Additive migration and adoption decision

12



OpenAPI supplies the invocation contract.

HAL affordances supply Resource-specific applicability and location.

x-iri-relation connects them without creating a second API contract.

Current proposal: 1 existing provisional relation + 24 registration requests. This RFC alone makes no production schema change.